MH1812 – Additional Exercises for Continuous Assessment 1

NTU, AY20/21 Semester 1

Q1: Prove that $\sqrt{3}$ is irrational.

Solution: Suppose that $\sqrt{3}$ is rational, that is, $\sqrt{3} = a/b$ for some integers a, b such that $\gcd(a, b) = 1$. Then

$$3b^2 = a^2. \tag{1}$$

There are two possible ways of proceeding at this point.

- Method 1: Since the left-hand side of (1) is divisible by 3, the right-hand side of (1) must also be divisible by 3, that is, a^2 is divisible by 3. Therefore a must be divisible by 3 (Check this yourself!). Hence we can write $a = 3k$ for some integer k. Plugging this back to (1) yields

$$3b^2 = (3k)^2,$$

or,

$$b^2 = 3k^2.$$

By a similar argument as above, b is divisible by 3. This contradicts our assumption that $\gcd(a, b) = 1$. Therefore, $\sqrt{3}$ is irrational.

- Method 2: If b is even, then the left-hand side of (1) will be even, so is the right-hand side of (1), which implies that a is even. This contradicts the assumption that $\gcd(a, b) = 1$. Therefore, b must be odd, and in this case, the left-hand side of (1) is odd and so is the right-hand side of (1), which implies that a is odd. We thus write $a = 2k + 1$ and $b = 2\ell + 1$ for some integers k and ℓ. Plugging them back to (1) yields

$$3(2\ell + 1)^2 = (2k + 1)^2,$$

or, after simplification,

$$6\ell^2 + 6\ell + 1 = 2(k^2 + k).$$

The left-hand side is odd but the right-hand side is even. We arrive at a contradiction. Therefore, $\sqrt{3}$ is irrational.

$\square$

Q2: Prove that $\sqrt{2}+\sqrt{3}$ is irrational.

Solution: We prove this by contradiction. If $\sqrt{2}+\sqrt{3}$ is rational, then $(\sqrt{2}+\sqrt{3})^2$ would also be rational. Note that

$$(\sqrt{2}+\sqrt{3})^2 = 2+3+2\sqrt{6} = 5+2\sqrt{6},$$

that is,

$$\sqrt{6} = \frac{(\sqrt{2}+\sqrt{3})^2-5}{2}$$

would be rational. However, we showed in the first lecture that $\sqrt{6}$ is irrational. Contradiction. Therefore, $\sqrt{2}+\sqrt{3}$ must be irrational. $\square$

Q3: Compute $5^{2015} \bmod 9$.

Solution: Observe that $5^6 \bmod 9 = 1$, thus

$$5^{2015} \bmod 9 = 5^{2015 \bmod 6} \bmod 9 = 5^5 \bmod 9 = 2.$$

$\square$

Q4: Compute $3^{3^{10}} \bmod 10$. [1]

Solution: Since $3^4 \bmod 10 = 1$, it holds that

$$3^{3^{10}} \bmod 10 = 3^{3^{10} \bmod 4} \bmod 10.$$

Now we calculate $3^{10} \bmod 4$. Observe that $3^2 \bmod 4 = 1$, we have that $3^{10} \bmod 4 = (3^2 \bmod 4)^5 \bmod 4 = 1$. Therefore

$$3^{3^{10}} \bmod 10 = 3^{3^{10} \bmod 4} \bmod 10 = 3^1 \bmod 10 = 3.$$

$\square$

Q5: Suppose a is an integer and p is a prime number such that $p|a$ and $p|(a+3)$. What can you deduce about p?

Solution: Since $p \mid a$ and $p \mid (a+3)$, we have $p \mid 3$. Therefore $p = 3$. $\square$

Q6: Prove by mathematical induction that, for $q \neq 1$ and integer $r \geq 0$,

$$a + aq + aq^2 + \cdots + aq^r = \frac{a(q^{r+1}-1)}{q-1}.$$

Solution: Let $P(k)$ denotes the proposition that the equation above holds for k.

- Base case: When $r = 0$, the left-hand side is a and the right-hand side is $\frac{a(q^{0+1}-1)}{q-1} = a$ and thus $P(0)$ is true.

[1] In an iterated exponential, one calculates the exponent first, $a^{b^c} = a^{(b^c)}$, e.g., $4^{3^2} = 4^9 = 262144$.

- Inductive step: Assume $P(k)$ is true: we want to show $P(k+1)$ is also true, that is, to show that

$$a + aq^1 + \cdots + aq^k + aq^{k+1} = \frac{a(q^{(k+1)+1} - 1)}{q-1} = \frac{a(q^{k+2} - 1)}{q-1}.$$

  Take the left-hand side:

$$\begin{aligned} a + aq^1 + \cdots + aq^k + aq^{k+1} &= \frac{a(q^{k+1} - 1)}{q-1} + aq^{k+1} \\ &= \frac{a(q^{k+1} - 1) + aq^{k+1}(q-1)}{q-1} \\ &= \frac{a(q^{k+2} - 1)}{q-1}, \end{aligned}$$

  where the first equality is due to the induction hypothesis. This completes the proof that $P(k+1)$ is true.

Therefore, by mathematical induction, $P(k)$ is true for all $k \geq 0$. $\square$

Q7: Let $a_k = a_1 + (k-1) \cdot d$ for non-negative integer k. Prove by mathematical induction that

$$\sum_{i=1}^{n} a_i = n \cdot \frac{a_1 + a_n}{2}.$$

Solution: Let $P(m)$ denotes the proposition that the equation above holds for m.

- Base case: $P(1) = \sum_{i=1}^{1} a_i = a_1 = 1 \cdot (a_1 + a_1)/2$
- Inductive step: Assume $P(m)$ is true: we want to show $P(m+1)$ is true, that is, to show that

$$P(m+1) = \sum_{i=1}^{m+1} a_i = (m+1) \cdot (a_1 + a_{m+1})/2$$

  Take the left-hand side:

$$\sum_{i=1}^{m+1} a_i = \sum_{i=1}^{m} a_i + a_{m+1} = m \cdot \frac{a_1 + a_m}{2} + a_{m+1} = \frac{m \cdot (a_1 + a_m) + 2a_{m+1}}{2}$$

  where the second equality is true because of the induction hypothesis.
  From the definition we have that

  1. $a_m = a_1 + (m-1) \cdot d \Longleftrightarrow ma_m = ma_1 + m(m-1)d$
  2. $a_{m+1} = a_1 + m \cdot d \Longleftrightarrow (m-1)a_{m+1} = (m-1)a_1 + m(m-1)d$

  Hence, $ma_m = (m-1)a_{m+1} + a_1$ Therefore

$$\begin{aligned} \sum_{i=1}^{m+1} a_i &= \sum_{i=1}^{m} a_i + a_{m+1} = m \cdot \frac{a_1 + a_m}{2} + a_{m+1} = \frac{m \cdot (a_1 + a_m) + 2a_{m+1}}{2} \\ &= \frac{ma_1 + (m-1)a_{m+1} + a_1 + 2a_{m+1}}{2} = (m+1) \cdot \frac{a_1 + a_{m+1}}{2} \end{aligned}$$

  This completes the proof of the inductive step.

Therefore, by mathematical induction, $P(m)$ is true for all $m \geq 1$. $\square$

Q8: Prove by mathematical induction that, for any integer $n \geq 1$,

$$(n+1)(n+2)\cdots(2n-1)(2n) = 2^n \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1),$$

Solution: Let $P(n)$ denote the proposition that the equation above holds for n.

- Base case: When $n = 1$, the left-hand side is $1 + 1 = 2$, the right hand side is $2^1 \cdot 1 = 2$, hence $P(1)$ is true.
- Inductive step: Assume $P(n)$ is true: we want to show $P(n+1)$ is also true. The left-hand side of $P(n+1)$ is

$$\begin{aligned}
&(n+2)(n+3)\cdots(2n)(2n+1)(2n+2) \\
&= (n+1)(n+2)\cdots(2n-1)(2n) \cdot \frac{(2n+1)(2n+2)}{n+1} \\
&= 2^n \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1) \cdot \frac{(2n+1)(2n+2)}{n+1} \\
&= 2^n \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1) \cdot (2n+1) \cdot 2 \\
&= 2^{n+1} \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1) \cdot (2n+1)
\end{aligned}$$

  which is the right-hand side of $P(n+1)$. Here we used the induction hypothesis that $P(n)$ is true for the second equality. This establishes $P(n+1)$.

Therefore, by mathematical induction, $P(n)$ is true for all $n \geq 1$. $\square$

Q9: Let p_n be the n-th prime number ($p_1 = 2$, $p_2 = 3$, $p_3 = 5$, ...). Show by mathematical induction that

$$p_{n+2} \geq 3n.$$

Solution: Let $P(n)$ denote the proposition that the inequality above is true for n.

- Base case: When $n = 1$, $p_{n+2} = 5 > 3n$, hence $P(1)$ is true.
- Inductive step: Suppose that $P(n)$ is true, that is, $p_{n+2} \geq 3n$. When $n > 1$, this implies that $p_{n+2} \geq 3n + 1$ since a prime cannot be $3n$ for $n > 1$; when $n = 1$, we also have $p_{n+2} \geq 3n + 1$. Hence we have $p_{n+2} \geq 3n + 1$ for all $n \geq 1$.

  We shall prove that $P(n+1)$ is true by contradiction. If $p_{n+3} < 3(n+1) = 3n+3$, since $p_{n+2} \geq 3n + 1$, it must be the case that $p_{n+2} = 3n + 1$ and $p_{n+3} = 3n + 2$. However, two consecutive integers cannot be both primes since one of them is even, so we have met a contradiction. Therefore, $p_{n+3} \geq 3(n+1)$, establishing $P(n+1)$.

Therefore, by mathematical induction, $P(n)$ is true for all $n \geq 1$. $\square$

Q10: Show by mathematical induction that

$$2 \cdot 7^n \equiv 2^n \cdot (5n + 2) \pmod{25}.$$

Solution: Let $P(n)$ denote the proposition that the equality above is true for n.

- Base case: When $n = 1$, LHS $= 2 \cdot 7 = 14$ and RHS $= 2^1 \cdot (5 + 2) = 14$, which are clearly congruent modulo 25.
- Inductive step: Suppose that $P(n)$ is true, we shall show that $P(n + 1)$ is true.

$$\begin{aligned}
2 \cdot 7^{n+1} \equiv 2 \cdot 7^n \cdot 7 &\equiv 2^n(5n + 2) \cdot 7 \quad \text{(induction hypothesis)} \\
&\equiv 2^n(35n + 14) \\
&\equiv 2^n(10n + 14) \\
&\equiv 2^{n+1}(5n + 7) \\
&\equiv 2^{n+1}(5(n + 1) + 2) \pmod{25}.
\end{aligned}$$

Therefore, by mathematical induction, $P(n)$ is true for all $n \geq 1$. □

Q11: Show $p \to (q \vee r) \equiv (p \wedge \neg q) \to r$

Solution:

$$\begin{aligned}
p \to (q \vee r) &\equiv \neg p \vee (q \vee r) \\
&\equiv (\neg p \vee q) \vee r \\
&\equiv (\neg p \vee \neg\neg q) \vee r \\
&\equiv (\neg(p \wedge \neg q)) \vee r \\
&\equiv (p \wedge \neg q) \to r
\end{aligned}$$

□

Q12: Show $p \to (q \to r) \equiv (p \wedge q) \to r$

Solution:

$$\begin{aligned}
p \to (q \to r) &\equiv \neg p \vee (q \to r) \\
&\equiv \neg p \vee (\neg q \vee r) \\
&\equiv (\neg p \vee \neg q) \vee r \\
&\equiv \neg(p \wedge q) \vee r \\
&\equiv (p \wedge q) \to r
\end{aligned}$$

□

Q13: Verify the following argument:

$$\begin{aligned}
&\neg p \vee q \to r \\
&s \vee \neg q \\
&\neg t \\
&p \to t \\
&\neg p \wedge r \to \neg s \\
\therefore\ &\neg q
\end{aligned}$$

Solution:

Step	Formula	Reason
(1)	$\neg p \vee q \to r$	Premise
(2)	$s \vee \neg q$	Premise
(3)	$\neg t$	Premise
(4)	$p \to t$	Premise
(5)	$\neg p \wedge r \to \neg s$	Premise
(6)	$\neg p$	(3)+(4), modus tollens
(7)	$\neg p \vee q$	(7), disjunctive addition
(8)	r	(1)+(7), modus ponens
(9)	$\neg p \wedge r$	(6)+(8), conjunctive addition
(10)	$\neg s$	(5)+(9), modus ponens
(11)	$\neg q$	(2)+(10), disjunctive syllogism

□

Q14: Verify the following argument:

$$\begin{aligned} & p \vee q \\ & q \to r \\ & p \wedge s \to t \\ & \neg r \\ & \neg q \to u \wedge s \\ \therefore\ & t \end{aligned}$$

Solution:

Step	Formula	Reason
(1)	$p \vee q$	Premise
(2)	$q \to r$	Premise
(3)	$p \wedge s \to t$	Premise
(4)	$\neg r$	Premise
(5)	$\neg q \to u \wedge s$	Premise
(6)	$\neg q$	(2)+(4), modus tollens
(7)	p	(1)+(6), disjunctive syllogism
(8)	$u \wedge s$	(5)+(6), modus ponens
(9)	s	(8), conjunctive simplification
(10)	$p \wedge s$	(7)+(9), conjunctive addition
(11)	t	(3)+(10), modus ponens

□

Q15: Verify the following argument:

$$\begin{aligned} &\neg p \to r \wedge \neg s \\ &t \to s \\ &u \to \neg p \\ &\neg w \\ &u \vee w \\ \therefore &\neg t \end{aligned}$$

Solution:

Step	Formula	Reason
(1)	$\neg p \to r \wedge \neg s$	Premise
(2)	$t \to s$	Premise
(3)	$u \to \neg p$	Premise
(4)	$\neg w$	Premise
(5)	$u \vee w$	Premise
(6)	u	(4)+(5), disjunctive syllogism
(7)	$\neg p$	(3)+(6), modus ponens
(8)	$r \wedge \neg s$	(1)+(7), modus ponens
(9)	$\neg s$	(8), conjunctive simplification
(10)	$\neg t$	(2)+(9), modus tollens

□

Q16: Determine whether the following statements are true or false.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x + y = 0$.
2. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R},\ x + y = 0$.

Solution:

1. The statement is true. Take $y = -x$.
2. The statement is false. For any x, let $y = -x + 1$, we have $x + y = 1 \neq 0$.

□

Q17: Write a new statement of the following by interchanging $\forall$ and $\exists$ (e.g., $\forall x \exists y$ becomes $\exists y \forall x$), and state the truth of both versions before and after the interchange.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x < y$.
2. $\exists x \in \mathbb{R},\ \forall y \in \mathbb{R}^-,\ x > y$. Here $\mathbb{R}^-$ denotes the set of negative reals.

Solution:

1. The original sentence is true. Take $y = x + 1$.
   If we interchange the $\forall$ and $\exists$, the sentence becomes

   $$\exists y \in \mathbb{R},\ \forall x \in \mathbb{R},\ x < y.$$

   This sentence is false. For any y, one can take $x = y + 1$, then $x < y$ does not hold
2. The original sentence is true. Take $x = 0$.
   If we interchange the $\forall$ and $\exists$, the sentence becomes

   $$\forall y \in \mathbb{R}^-,\ \exists x \in \mathbb{R},\ x > y.$$

   This sentence is true. Take $x = 0$.

$\square$

Q18: Write the negation of the following statements.

1. For every real number x, if $x^2 \geq 1$ then $x > 0$.
2. For any integers a, b and c, if $a - b$ is even and $b - c$ is even, then $a - c$ is even.

Solution: Note that $\neg(p \to q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q$.

1. There exists a real number x such that $x^2 \geq 1$ and $x \leq 0$.
2. There exist integers a, b, c such that $a - b$ and $b - c$ are even while $a - c$ is odd.

$\square$

Q19: Write the negation of the following sentences.

1. $\forall x \in D, \forall y \in E, P(x, y)$
2. $\exists x \in D, \exists y \in E, P(x, y)$
3. $\exists x \in D, \forall y \in E, \exists z \in F, \neg P(x, y, z) \to Q(x, y, z)$

Solution:

1. $\exists x \in D, \exists y \in E, \neg P(x, y)$
2. $\forall x \in D, \forall y \in E, \neg P(x, y)$
3. $\forall x \in D, \exists y \in E, \forall z \in F, \neg P(x, y, z) \wedge \neg Q(x, y, z)$.

$\square$